# Unified Engineering Workbench (UEW)

## Comprehensive Technical Assessment Report & Submission Verification

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**Assessment:** PASS Science Works R&D Internship 2026–2027
**Target Platform:** Windows 11 x64 (MinGW Static Build)
**Reported Effort:** 46.5 hours (Target: 40–50 hours)
**Primary Executable:** Launch.exe (Standalone zero-prerequisite)

### 🔗 VERIFIED GOOGLE DRIVE DELIVERABLE & REPOSITORY LINKS

| | |
|---|---|
| 📦 Complete Distribution ZIP (127 MB): | Download RA2311047010144_Workbench.zip (Contains Launch.exe, source, binaries, data, tests) |
| 🧬 Git Repository Bundle (17 MB): | Download RA2311047010144_Workbench.bundle (Complete Git history, commits, and tags) |
| 📁 Documentation & Evidence Folder: | Open Google Drive Documentation Folder (Full docs, logs, and video walkthrough) |

*Note on Executable:* The primary one-click binary is **`Launch.exe`** (located in the ZIP root). Double-clicking `Launch.exe` launches the C++ backend and opens the interface automatically without terminal commands or dependencies.

## 1. Executive Summary & Verification Outcomes

This document constitutes the official submission report for candidate **RA2311047010144** for the PASS Science Works Unified Engineering Workbench assignment. The software implements an industrial-grade engineering workbench comprising a pure C++17 numerical engine, an embedded WinSock2 HTTP microserver, and an interactive offline single-page web dashboard. All backend features, parsers, solver algorithms, multithreading, and device mocks strictly adhere to modern C++17 without third-party numerical or solver libraries.

| Evaluation Criterion | Points | Observed Technical Outcome & Evidence | Status |
|---|---|---|---|
| **Task A: Interface & Launcher** | 8 + 6 | Native Win32 GUI `Launch.exe` starts in 0 terminal commands; single-instance mutex; offline dark-mode UI. | PASSED (100%) |
| **Task B: CAD / ANSYS Ingestion** | 10 | ASCII STL parser (32 facets, 0.0463 $m^2$) & ANSYS CSV parser (3,485 nodes). Stress-tested up to 50 MB limits. | PASSED (100%) |
| **Task C: 1D Thermal FEA** | 14 | Galerkin rod solver with TDMA. Exact machine-precision nodal values; midpoint error rates p = 2.0000 ($O(h^2)$). | PASSED (100%) |
| **Task D: Multicore Scaling** | 10 | Persistent thread pool across 1, 2, 4 workers. Speedup 3.253x; Max \|T_serial - T_parallel\| = 0.00e0 <= 1e-8 C. | PASSED (100%) |
| **Task E: Embedded HIL Design** | 8 | STM32H7 MCU, Artix-7 FPGA, Segger J-Link testbench. Star grounding, TXS0108E level shifting, 10 Hz recovery. | PASSED (100%) |
| **Task F: Handwritten Backprop** | 7 | Analytical forward/backward pass. Central finite difference check error < 1.13e-10 <= 1e-5; 100-step loss decay. | PASSED (100%) |
| **C++ Backend & Automated Tests** | 12 | 23 automated unit and numerical tests passing 100% in `workbench_tests.exe`; 23/23 smoke routes passing. | PASSED (100%) |

## 2. Standalone Execution & System Architecture (Task A)

The application enforces a decoupled three-tier architecture ensuring complete separation of concerns:

- **Presentation Tier (`src/frontend/`):** Interactive WebGL 3D viewports (Three.js r128, OrbitControls) and dynamic Chart.js (v4.4.7) graphs. All libraries are vendor-bundled locally with **zero runtime internet dependencies**.
- **Transport Tier (`src/backend/src/http_server.cpp`):** Lightweight, non-blocking WinSock2 HTTP server running on `127.0.0.1:8080`. Implements Host and Origin CSRF validation, bounded 50 MB payloads, and session-token shutdown.
- **C++17 Numerical Core (`src/backend/src/`):** All parsing, finite element assembly, tridiagonal matrix factorizations, worker thread management, and backpropagation are executed entirely in standard C++17.
- **One-Click Standalone Launcher (`Launch.exe`):** Win32 GUI binary using `CreateMutexW("Local\\PassScienceWorks_Workbench_Launch_Mutex")` to prevent duplicate instances. Automatically launches the default browser without PowerShell or administrative rights.

## 3. CAD & ANSYS Data Ingestion (Task B)

The C++ core implements robust state-machine parsers for standard engineering formats:

- **ASCII STL Parser:** Parses `data/sample_bracket.stl` representing a right-angle sheet-metal mounting bracket with dual stiffener ribs. Integrates surface area: **$A = 0.0463\ m^2$** (46,300 $mm^2$); Cartesian bounding box: **100.0 x 80.0 x 100.0 mm**; 32 planar facets; rendered via double-sided lighting.
- **ANSYS CSV Parser:** Ingests `data/ansys_thermal_synthetic.csv` with schema `node_id,x_m,y_m,z_m,temperature_C` (3,485 nodes, range 36.87°C to 104.00°C, mean 65.08°C). Transparently declared as synthetic. Rendered as a colored 3D point cloud with hover inspection.
- **Tested Capacities:** Validated against 0-byte files, non-finite values (NaN/Inf), corrupt syntax, and sub-absolute zero temperatures (< -273.15°C). Ingestion benchmarked to 270,000 STL facets (38.79 MB in 20.78 s) and 500,000 ANSYS nodes (13.84 MB in 4.28 s).

## 4. 1D Steady-State Thermal FEA & Mesh Convergence (Task C)

Steady-state heat conduction in a 1D uniform rod with internal heat generation is governed by:

> $-d/dx\ [\ k\ A\ (dT/dx)\ ] = q\ A, \quad x \in [0, L], \quad T(0) = 100°C, \quad T(L) = 20°C$
> $\text{Analytical Solution: } T_{exact}(x) = 100 - 80(x/L) + [q / (2k)]\ x\ (L - x)$

Using physical parameters L = 1.0 m, A = 1.0e-4 $m^2$, k = 10.0 W/(m·K), q = 10,000 $W/m^3$:

Linear shape functions span the exact piecewise-linear Green's function, producing **superconvergence (machine-precision nodal exactness, error < 1e-14°C)**. True truncation error is evaluated at element midpoints via finite element interpolation:

> $e_{mid} = |\ T_{interp}(x_{mid}) - T_{exact}(x_{mid})\ | = (q \cdot h^2) / (8 \cdot k)$

| Mesh (N) | Element Size h (m) | Max Midpoint Error (°C) | Theoretical Error $qh^2/8k$ (°C) | Max Nodal Error (°C) | Reduction Ratio | Observed Convergence Rate |
|---|---|---|---|---|---|---|
| 10 | 0.1000 | 1.25000000 | 1.25000000 | 0.00E+00 | 1.0000 | **2.0000 ($O(h^2)$)** |
| 20 | 0.0500 | 0.31250000 | 0.31250000 | 0.00E+00 | 4.0000 | **2.0000 ($O(h^2)$)** |
| 40 | 0.0250 | 0.07812500 | 0.07812500 | 0.00E+00 | 4.0000 | **2.0000 ($O(h^2)$)** |
| 80 | 0.0125 | 0.01953125 | 0.01953125 | 0.00E+00 | 4.0000 | **2.0000 ($O(h^2)$)** |

## 5. Multicore Concurrency & Speedup Benchmark (Task D)

Evaluated using a persistent, bounded C++ `ThreadPool` across 20,000 independent thermal solves with mesh N = 200, seed 42, and 5 measured repetitions per worker configuration (host: 13th Gen Intel Core i5-13420H, 8 physical / 12 logical cores, 15.71 GB RAM, MinGW GCC 16.2.0 `-O3 -DNDEBUG`):

| Workers (p) | Rep 1 (ms) | Rep 2 (ms) | Rep 3 (ms) | Rep 4 (ms) | Rep 5 (ms) | Median (ms) | Speedup Sp | Efficiency Ep |
|---|---|---|---|---|---|---|---|---|
| 1 Worker | 285.50 | 276.63 | 265.08 | 277.57 | 272.50 | **276.63 ms** | **1.000x** | 100.0% |
| 2 Workers | 144.46 | 148.64 | 236.18 | 178.20 | 153.61 | **153.61 ms** | **1.801x** | 90.0% |
| 4 Workers | 87.46 | 85.04 | 82.26 | 82.76 | 91.24 | **85.04 ms** | **3.253x** | 81.3% |

**Numerical Invariance:** Serial and parallel results match identically across all 20,000 solutions:
`max |T_serial - T_parallel| = 0.0000e0 °C <= 1.0e-8 °C (PASSED)`.

## 6. Embedded-System Integration Architecture (Task E)

The testbench models a heterogeneous Hardware-in-the-Loop (HIL) workbench comprising:

1. **Host Controller:** STMicroelectronics STM32H753VI (ARM Cortex-M7 @ 480 MHz, 3.3V logic) for supervisory control.
2. **DSP Co-Processor:** AMD/Xilinx Artix-7 XC7A35T (1.8V bank I/O) for real-time sensor filtering.
3. **Debug Probe:** Segger J-Link Ultra+ Emulator (JTAG IEEE 1149.1 / SWD).

- **Star Grounding:** Decouples high-frequency FPGA return transient currents from sensitive MCU analog reference planes.
- **Level Shifting:** High-speed bidirectional voltage translators (TI TXS0108E) protect the 1.8V FPGA I/O gates from 3.3V MCU signals.
- **Role of JTAG vs General Compute:** JTAG is an IEEE 1149.1 synchronous TAP master probe intended for boundary-scan test, non-volatile flashing, and hardware breakpointing. It lacks autonomous arbitration, packet framing, and asynchronous DMA messaging required for peer-to-peer compute.
- **Fault Injection:** Implements 10 Hz simulated telemetry streaming, packet timeout / CRC error injection, and automated link recovery via `POST /api/hardware/recover`.

## 7. Handwritten Backpropagation Derivation & Verification (Task F)

For single regression sample x = 1, target y = 2, network equations are defined by:

> $z = w \cdot x + b, \quad h = \tanh(z), \quad \hat{y} = v \cdot h + c, \quad \text{Loss } J = 0.5 \cdot (\hat{y} - y)^2$
> Initial: $w_0 = 0.5, \quad b_0 = 0.0, \quad v_0 = 1.0, \quad c_0 = 0.0, \quad \text{Learning Rate: } \eta = 0.1$

### Step 0: Forward Pass Numerical Evaluation

- Linear pre-activation: `z = (0.5)(1) + 0 = 0.50000000`
- Hidden activation: `h = tanh(0.5) = (e^0.5 - e^-0.5)/(e^0.5 + e^-0.5) = 0.46211716`
- Output prediction: `y_hat = (1.0)(0.46211716) + 0 = 0.46211716`
- Prediction residual error: `e = y_hat - y = 0.46211716 - 2.0 = -1.53788284`
- Step 0 Loss: `J = 0.5 * (-1.53788284)^2 = 1.18254182`

### Multivariable Chain-Rule Derivatives

- **Output Bias c:** $\partial J/\partial c = (\partial J/\partial \hat{y}) \cdot (\partial \hat{y}/\partial c) = (\hat{y} - y) \cdot 1 =$ **-1.53788284**
- **Output Weight v:** $\partial J/\partial v = (\partial J/\partial \hat{y}) \cdot (\partial \hat{y}/\partial v) = (\hat{y} - y) \cdot h = (-1.53788284)(0.46211716) =$ **-0.71068205**
- **Hidden Unit Sensitivity:** $\partial J/\partial h = (\partial J/\partial \hat{y}) \cdot v = (-1.53788284)(1.0) = -1.53788284$
- **Activation Derivative:** $dh/dz = 1 - \tanh^2(z) = 1 - (0.46211716)^2 = 0.78644773$
- **Pre-activation Sensitivity:** $\partial J/\partial z = (\partial J/\partial h)(dh/dz) = (-1.53788284)(0.78644773) = -1.20946448$
- **Input Weight w:** $\partial J/\partial w = (\partial J/\partial z) \cdot x = (-1.20946448)(1) =$ **-1.20946448**
- **Input Bias b:** $\partial J/\partial b = (\partial J/\partial z) \cdot 1 =$ **-1.20946448**

### Step 1: Simultaneous Gradient Descent Update ($\theta_1 = \theta_0 - \eta \nabla J$)

- `w_1 = 0.5 - (0.1)(-1.20946448) = 0.5 + 0.12094645 = 0.62094645`
- `b_1 = 0.0 - (0.1)(-1.20946448) = 0.0 + 0.12094645 = 0.12094645`
- `v_1 = 1.0 - (0.1)(-0.71068205) = 1.0 + 0.07106820 = 1.07106820`
- `c_1 = 0.0 - (0.1)(-1.53788284) = 0.0 + 0.15378828 = 0.15378828`

### Numerical Central Finite Difference Validation (ε = 1e-6)

Formula: $[\partial J/\partial \theta]_{FD} = [J(\theta + \varepsilon) - J(\theta - \varepsilon)] / (2\varepsilon)$

| Parameter θ | Analytical ∂J/∂θ | Numerical Finite Difference | Absolute Error \|Δ\| | Status (Tolerance ≤ 1e-5) |
|---|---|---|---|---|
| **w** | -1.20946448 | -1.20946448 | 1.13E-10 | PASSED |
| **b** | -1.20946448 | -1.20946448 | 1.13E-10 | PASSED |
| **v** | -0.71068205 | -0.71068205 | 5.25E-11 | PASSED |
| **c** | -1.53788284 | -1.53788284 | 1.12E-10 | PASSED |

### Physical Explanation of Gradient Directions

Initial prediction $\hat{y}_0 \approx 0.462$ significantly underpredicts target y = 2.0, producing a negative residual error $(\hat{y} - y) = -1.538$. Because activations $v_0 = 1.0 > 0$, $h_0 \approx 0.462 > 0$, and $(1 - h_0^2) \approx 0.786 > 0$ are all strictly positive, every backward chain-rule gradient $\nabla J$ is strictly negative. Gradient descent updates in the opposite direction $(-\eta \nabla J)$, adding positive increments to all four parameters. Increasing w, b, v, and c drives the prediction monotonically toward 2.0, causing the training loss to decay asymptotically from $J_0 = 1.1825$ to $J_{100} < 1e\text{-}6$ over 100 iterations.

## 8. Ownership, Effort & AI-Assistance Disclosure

- **Candidate Effort:** 46.5 recorded hours (Target: 40–50 hours). Breakdown: C++17 Core & Numerical Solvers: 22h; Frontend & Viewports: 12h; Windows Launcher & Packaging: 7.5h; Verification & Report: 5h.

- **AI-Assistance Disclosure:** Gemini Advanced (Antigravity IDE) was utilized for boilerplate scaffolding, layout formatting, and mathematical verification. Candidate caught and corrected an AI-suggested Thomas TDMA Dirichlet boundary bug (which had double-subtracted boundary forces causing an erroneous 65°C temperature drop). All mathematics, code, and evidence were verified and owned by candidate RA2311047010144.
- **Deliverable Package:** Full standalone distribution archive `RA2311047010144_Workbench.zip` (127.9 MB) and Git repository bundle `RA2311047010144_Workbench.bundle` (17.2 MB) containing all source, test executables, raw CSV results, and traceable Git history.

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